SOFTWARE REQUIREMENTS SPECIFICATION

VSM

DIGITAL SERVICES PLATFORM

Multi-Service Digital Ecosystem

A unified platform combining loan management and photography booking with an integrated financial ecosystem.

💳 Loan Management System

📷 Photography Booking System

PREPARED BY

TD Software Solutions

DATE

April 2026

VERSION

1.0

SECTION 01

Platform Strategy

PRODUCT VISION

One Platform. Multiple Services.

The VSM Digital Services Platform combines a Loan Management System and a Photography Booking System into a single unified platform — modularly separated for clarity, yet sharing core infrastructure for seamless operations.

🏗️ MODULAR ARCHITECTURE

- VSM Platform → Loan Services
- VSM Platform → Photography Services
- Shared: Users, Payments, Notifications

💡 UNIQUE VALUE PROPOSITION

- Select a photography package
- Choose "Pay with Loan"
- Auto-apply and get approved
- Closed-loop financial ecosystem

2	1	4
Services on Platform	Integration Feature	Dev Phases

👥 FOR CUSTOMERS

- Access services + financing in one place
- Convenience and faster processing

🏢 FOR BUSINESS

- Increased revenue streams
- Higher customer conversion
- Competitive market advantage

SECTION 02

System Overview

🌍 PLATFORM TYPE

👤 CUSTOMER ROLE

- Apply for loans

🛡️ ADMIN ROLE

- Manage loans & bookings

Web-Based

Multi-Service

Role-Based

- Book photography services

- Monitor the system

SECTION 03

Loan Management System

📈

10%

Fixed Interest Rate

💰

R100–R10K

Loan Range

📅

1 Mo

Repayment Period

👤 CUSTOMER FEATURES

- Register & login securely
- Apply for loans online
- Upload required documents
- View live loan status
- View repayment schedule
- Access loan history dashboard

💳 PAYMENT METHODS

- Debit Order **Priority**
- Bank Transfer
- PayFast
- Ozow

🧠 SMART FEATURES

- Loan Eligibility Checker (income-based)
- Credit Scoring System (income + repayment behavior)

⚠️ LOAN LOGIC

- Manual approval (admin reviews)
- Automatic approval (rule-based)
- Late payment penalty system

🔔 NOTIFICATIONS

📧 Email 📱 SMS 💬 WhatsApp

🛡️ ADMIN FEATURES

- View & approve/reject loans
- Risk dashboard: loans, defaults, revenue

🚨 FRAUD DETECTION

- Duplicate ID detection
- Suspicious activity patterns

📄 LOAN AGREEMENT SYSTEM

Auto-Generated PDF | Digital Signing | T&C Included

SECTION 04

Photography Booking System

👤 CUSTOMER FEATURES

- Browse photography packages
- Book events with date selection
- Make payments online
- Access private client gallery
- Download photos privately

📅 BOOKING & CALENDAR

- Date selection interface
- Real-time availability checking
- Booking confirmation system
- Double-booking prevention
- Upcoming events calendar

📦 PACKAGES SYSTEM

Basic Package | Standard Package | Premium Package

🛡️ ADMIN FEATURES

- Manage bookings & packages
- Track payments
- View upcoming events

SECTION 05

System Integration — Finance Your Event

CLOSED-LOOP FINANCIAL ECOSYSTEM

"Finance Your Event" Feature

📷 Select Package → 💳 Pay with Loan → 📝 Auto-Fill Application → ✅ Loan Approved → 🔗 Payment Linked

SECTION 06

Shared System Components

🔐 AUTHENTICATION

- Secure Login
- Role-Based Access

💳 PAYMENTS

- Shared Gateway
- Loans & Bookings

🔔 NOTIFICATIONS

- Email
- SMS
- WhatsApp

SECTION 07

System Architecture

🖥️ FRONTEND

- HTML
- CSS
- JavaScript
- GSAP
- AOS

⚙️ BACKEND

- Python
- Flask

🗄️ DATABASE

- PostgreSQL

☁️ HOSTING

- Vercel

SECTION 08

Database Design

👤 USERS

id
name
email
password
role

💰 LOANS

id
user_id
amount
interest
status

💳 PAYMENTS

id
type (loan/booking)
amount
status

📷 BOOKINGS

id
user_id
package_id
event_date

📦 PACKAGES

id
name
price

📁 DOCUMENTS

id
user_id
file_url

SECTION 09

Security Requirements

🌐

🔑

✅

HTTPS

End-to-end encryption

Auth

Secure authentication

Validation

Input & data validation

File Protection

Secure upload pipeline

Fraud Detection

Automated anomaly detection

SECTION 10

UI / UX Requirements

Modern	Animated	Responsive	Dashboard
Clean Interface	Smooth Transitions	Mobile Friendly	Style Layouts

SECTION 11

Development Plan

1 Phase 1 — MVP Foundation

- Loan System Core
- Authentication
- Admin Dashboard

2 Phase 2 — Enhanced Loan Features Growth

- Payment Integrations
- Notifications
- Loan Automation

3 Phase 3 — Photography System Expansion

- Photography Booking
- Gallery System

4 Phase 4 — Full Integration Complete

- Finance Your Event Feature
- Closed-Loop Ecosystem

SECTION 12

Future Enhancements

🤖

AI Approval

Smart loan decisions

📱

Mobile App

iOS & Android

📊

Analytics

Advanced insights

⭐

Credit Score

Behavioral scoring

Section 13 — Conclusion

The **VSM Digital Services Platform** is a scalable, multi-service system designed to digitize loan processing, enable seamless photography bookings, and integrate financial services with real-world customer needs — all within a single, powerful digital ecosystem.

END OF DOCUMENT